

the program with command line arguments. It shows a text box where the command line arguments are to be entered, each separated by a space.

C4droid is a very handy tool for education and programming, while on the go.

Frink

Frink is a calculating tool and programming language designed by Alan Eliassen. It is built on the Java Virtual Machine and incorporates features similar to Java, Perl, Ruby, Smalltalk and various BASIC implementations. Its main focus is in the fields of science, engineering, physics, text processing and education.

Installation

Frink for Android is available on Google Play. Search for 'Frink Programming Language'. Frink is a free application in Google Play and is also available for use on your PC. Versions for Linux, Mac and Windows are available.

Features

On systems with an Internet connection, Frink can translate dozens of human languages, including some requiring right-to-left reading; it can also convert current currencies and calculate precious metal values as well as inflation-adjusted US dollar and British pound values.

Some features specifically available in the Android version are:

- Interactive mode - This mode allows interactive calculations, making it a powerful calculating tool.
 - Programming mode - Allows writing, saving and loading full-fledged Frink programs.
 - Graphics - All of Frink's graphics features are available, including transparent, anti-aliased graphics. Graphics can be written to PNG, JPEG and SVG formats.
 - Images - Images can be loaded, modified, displayed and written out to the device's internal storage.
 - Animation - Graphics can be animated using Frink's animation capabilities.
 - Sensors - All of the sensors on Android can be read.
 - GPS - Allows fetching of a location from the GPS receiver or from the network-based location services. The GPS receiver can be turned on and off by programming it. You can access either the last-known position or turn on the GPS function and acquire current, continuously-updated position information.
 - Text-to-speech - Frink can easily speak text out aloud using a single `speak[string]` function.
 - Speech recognition - Frink can turn your voice input into text using a single `listen[prompt]` function. It can then translate that text into other languages, speak it aloud and perform calculations with it!
- Apart from these main features, the Android version of Frink has a programmer-friendly interface. It provides

on-screen keys that make programming easier, while parentheses, curly braces and double quotes can be typed in pairs. This is done because typing large programs on small devices can be really tiring.

Frink can also be used as a calculator and as an app for unit conversions.

JoyToy programming

This is yet another programming language available in Android. Of the three mentioned here, this is the simplest and least complicated IDE. It was originally written for the book "Imadoki no program gengo no tukurikata" in Japan. Perhaps it has the simplest documentation available.

Installation

Just like C4droid and Frink, JoyToy is also available on Google Play. It's just a 60.5 KB app. Search for it and you can download and install it.

Features

The JoyToy interface has an edit panel to write source code and a console panel to the right of the edit panel where the results are displayed. At the bottom of the screen are five buttons with the following functions:

1. Eval - to run the program
2. Load - displays a files list to choose an already saved program
3. Save - to save the current file
4. Clear - to clear the console panel
5. Help - to display help

JoyToy has the 'if' expressions and also has the 'while' loop. For now, other loops are not available on JoyToy. It has a limited set of built-in functions also. But you can define variables and functions like in any other language.

JoyToy also has object-oriented features. You can create objects, member variables and the functions associated with them.

There are a lot more features for these programs. I expect readers to download and try these programs and experience the joy of programming on Android rather than programming for Android devices. 

References

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